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Lab to Field: Technology Demonstration of Multi-Species (CO₂, SO₂, NO₂, & PM) Capture from a Power Plant Exhaust

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Abstract

Post-combustion capture of carbon dioxide usually requires other upstream pollutant capture systems such as selective catalytic reduction for NO_x, flue gas desulfurization, electrostatic precipitators, etc. Traditional carbon capture technologies cannot be deployed at point emission sources that do not have these systems and use fuels containing sulfur (e.g., heavy fuel oil, sour gas, etc.). A novel pollutant capture system is being developed at KAUST that eliminates this limitation.

A Cryogenic Carbon and Sulfur Capture (CCSC) technology has been developed at KAUST in collaboration with the Saudi Electricity Company (SEC). This technology focuses on post-combustion CO₂ capture along with SO₂, NO₂, and other Particulate Matter (PM) pollutants. CCSC is modular in design and mounted on a 15-meter-long trailer. The multi-species co-capture system is mounted in the central section of the trailer and includes a separate control room and gas storage room on either end of the trailer. This trailer-mounted CCSC system has a nominal capacity of capturing 0.25 ton of CO₂/day.

This CCSC process cools exhaust flue gases below the desublimating temperatures of CO₂ where CO₂ solidifies out of the flue gas. This separation process is done by bubbling the flue gas through the pool of isopentane at cryogenic temperatures, which serves as a contact liquid. CO₂ freezes into the contact liquid and flows out as a slurry. The contact liquid is regenerated via a mechanical solid-liquid separator, and then the CO₂ product is recovered via distillation. An additional column is necessary for the separation of a third species SO₂.

The main objective of this mobile CCSC rig is to serve as a technology and IP development platform to co-capture multiple pollutants, all using a single technology. The trailer rig is designed to be flue gas agnostic and finds applications in a wide range of industries. Even though the system is currently optimized for a heavy fuel oil (HFO) fired steam power plant, it can easily be adapted to shipboard carbon capture and co-capture of other pollutants more prominent from marine applications. A wetted-wire patented technology from KAUST already shows promise for shipboard heat and mass transfer. This paper presents a multipollutant capture demonstration from the exhaust of an operational power plant, validating the system's performance under real-world conditions.

Introduction

Cryogenic Carbon and Sulfur Capture (CCSC) is an emerging post-combustion carbon capture technology that separates CO₂ by cooling flue gas to cryogenic temperatures where CO₂, along with other pollutants, solidifies, enabling its separation from other gas components. Figure 1 shows the simplified process flow set up for CCSC process on the trailer rig. Unlike solvent-based systems, CCSC uses phase change principles and typically operates at cryogenic temperatures around $\sim -130^{\circ}\text{C}$, eliminating the need for chemical absorbents and reducing regeneration energy penalties. The process can achieve high CO₂ purity and capture efficiency while being retrofittable to existing infrastructure with an optimal footprint that is modular. CCSC is particularly promising for applications requiring low water usage or integration with downstream liquefaction or utilization processes. The Cryogenic Carbon Capture (CCC) process has been described as a carbon capture technology with among the greatest potential for impact (Herzog, 2018). Other cryogenic carbon capture approaches have also been presented by Baxter et al. (2021), Frankman et al. (2021), and Hoeger et al. (2021), who have provided additional information on the process and its underlying principles, and more generally in reviews by Song et al. (2019) and by Font-Palma et al. (2021).

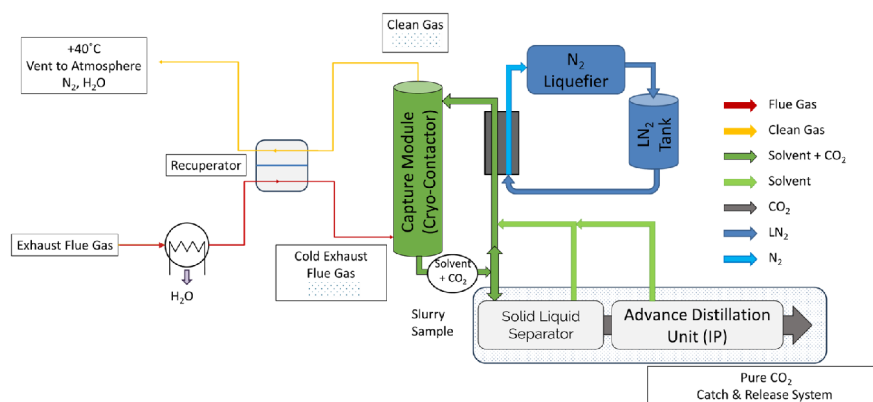


Figure 1—Simplified diagram of the CCSC process on the trailer rig

While amine-based systems are regarded as the mature and preferred option for CO₂ capture in chemical processing and fuel production, owing to their ability to integrate with process heat and their established operational reliability, CCSC is considered a promising alternative for the power generation sector, particularly in addressing point-source emissions containing multiple pollutants. In CCSC, pollutants along with CO₂ are captured as frozen solids, which are subsequently melted and pumped in liquid form to high pressures (up to 150 bar), resulting in significantly lower energy consumption compared to the gas-phase compression required by conventional systems. As the process does not require heat integration or solvent regeneration, it is suited for bolt-on retrofit applications. Given these advantages, CCSC is viewed as a viable solution not only for power plants but also for marine and other hard-to-abate sectors. CO₂ capture efficiencies exceeding 98% have been demonstrated and are detailed within this paper, highlighting the performance potential of the Cryogenic Carbon Capture system.

Among cryogenic processes, both the CCSC and the Advanced Cryogenic Carbon Capture (A3C) systems, as described by Wilson (2020), utilize a circulating contact medium. In the A3C process, solid steel balls are used as the heat transfer medium instead of a liquid contactor, resulting in a substantial reduction in desublimator vessel size. This compact design has led to its proposal for marine applications, where space constraints are critical. However, a key limitation of the A3C process is that it produces CO₂ in gaseous form, thereby requiring additional compression to reach transport-ready conditions. Further energy losses may arise from the cooling method, in which steel balls are chilled via direct contact with a refrigerant,

most likely liquid nitrogen, operating in an open-loop configuration with the atmosphere. Although the A3C process offers a smaller footprint, it is considered more energy-intensive than the CCSC system.

The objective of the first-of-its-kind Carbon Capture and Sulphur Capture (CCSC) rig is to enable the efficient separation of CO₂, SO₂, NO₂, and particulate matter from flue gas streams. This report presents the performance and separation efficiency of KAUST's 0.25 tonne-per-day CCSC trailer rig. The rig was commissioned at the largest power plant operated by the Saudi Electricity Company, and the outcomes of this field deployment are summarized herein. The CCSC system is fuel-agnostic and designed for ease of retrofit into existing utility and heavy-emitting industrial facilities. Conventional emission control technologies, such as Flue Gas Desulfurization (FGD), Selective Catalytic Reduction (SCR), and Electrostatic Precipitators (ESP), are typically deployed to target SO_x, NO_x, and particulate matter separately. These systems not only contribute to higher capital expenditures but also impose substantial auxiliary power demands on the host plant. By consolidating the functions of FGD and SCR into a single downstream CCSC unit, significant reductions in both CAPEX and OPEX, along with lower energy consumption, can be realized.

Technology & Process Description

System Architecture & Design Philosophy

The Cryogenic Carbon and Sulphur Capture (CCSC) rig operates as a catch-and-release system ([Figure 1](#)), serving both as an intellectual property (IP) demonstration platform and a test facility for evaluating novel carbon capture technologies. Flue gases are drawn directly from the point source of emission and subjected to cryogenic cooling, resulting in the condensation and solidification of CO₂ along with co-captured pollutants such as SO_x and NO_x. The treated, clean air is safely vented to the atmosphere, while a slurry comprising the liquid coolant (iso-pentane) and solid CO₂ is separated and subsequently pressurized and routed to a distillation stage. The iso-pentane is recovered and recirculated into the cooling loop, while the captured CO₂ is either released or stored in a dedicated sampling vessel for validation of CO₂ purity. The purity of the sampled CO₂ is measured using Fourier Transform Infrared Spectroscopy (FTIR).

[Figure 2](#) shows a representative CAD drawing of the trailer-mounted CCSC rig. The trailer is organized into four distinct compartments, each physically separated by fire-resistant partition walls to ensure safe and efficient operation. These compartments are shown in [Figure 3](#) and include:

1. **Control Room:** Houses Supervisory Control and Data Acquisition (SCADA), monitoring systems, and operator interfaces.
2. **Process Operational Zone:** Contains key process equipment and supports the main gas handling operations.
3. **Cryo Generator Room:** Encloses the cryogenic refrigeration unit used for low-temperature operation.
4. **Air-Conditioned Chemical Storage:** Designated for the safe storage of solvents, refrigerants, and other process-related chemicals.

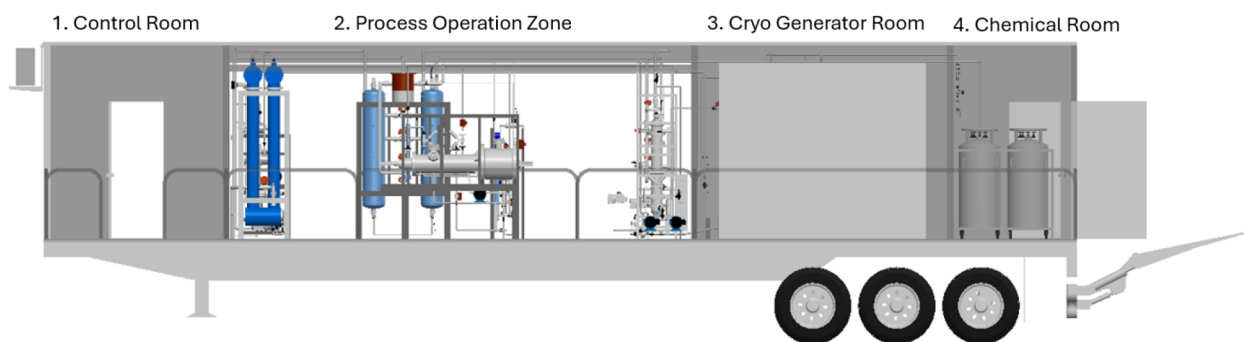


Figure 2—Representative 3D Model of the Trailer Rig (Plan View)

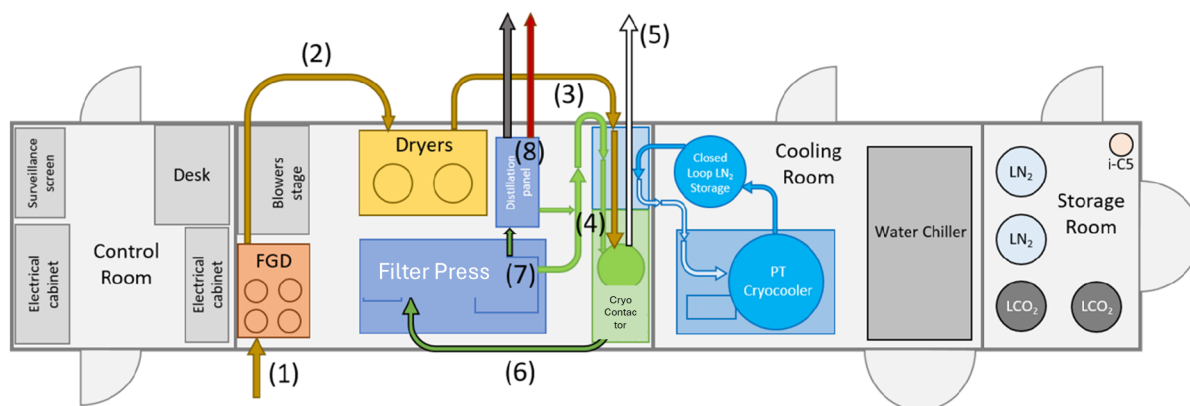


Figure 3—CCSC system layout on the flatbed trailer rig (1) Flue Gas Desulphurization (2) Dryers (3) Recuperator (4) Cryo-Contactor (5) Clean Gas Exit (6) Filter Press (7) Pollutant Melter Pump (8) Desublimation Panel

Within the Process Operational Area, the following major subsystems/modules are installed:

- **Flue Gas Conditioning Module:** Ensures gas flow assurance, dehydrates, and removes bulk particulate matter through pre-filtration.
- **Heat Exchanger Integration Module:** Facilitates thermal coupling between incoming flue gas and the refrigeration cycle.
- **Capture/Separation/Purification Module:** Performs the core function of cryogenic capture, separation, and purification of CO₂.

Based on the operating conditions, modules are enclosed within a customized, state-of-the-art cold box to minimize thermal losses to the ambient environment and ensure efficient cryogenic operation.

Figure 3 The layout of the major equipment on the flatbed trailer. Arrows indicate the flue gas path within the system.

Process Flow and Operating Principles

The exhaust flue gas from the point source, with CO₂ concentration in the range of 4%-15%, enters the FGD (marked in orange) through the quick-connect flexible hose under the trailer. This flue gas entering the system generally has water vapor and SO₂ among other pollutants. The FGD is in place to control the incoming SO₂ concentration to be maintained at or below 2000 ppm. The FGD also allows for tuning the concentration of SO₂ as per design basis requirements (**Figure 3**).

The flue gas then passes through a series of aftercoolers which cool the flue gas further. This wet flue gas with an SO₂ concentration enters an alternating dryer. The dryers are packed with molecular sieves along with regeneration heaters in series, which condense the water vapor from the flue gas and ensure that the flue gas exiting the system is dry.

The dry flue gas then enters the recuperative heat exchanger before entering the Cryocontactor (marked in green). The flue gas in the recuperator cools down by the cold outgoing clean gas from the Cryocontactor. This recuperator makes the system efficient and decreases any additional energy that might be needed for cooling the incoming flue gas. The refrigeration cooling loop condenses gaseous nitrogen to liquid nitrogen, which in turn cools the contact liquid down to cryogenic temperatures $< -130^{\circ}\text{C}$.

The flue gases are bubbled through the pool of contact liquid, which is at cryogenic temperatures, where CO_2 and SO_2 desublimates into the contact liquid and flow out of the Cryocontactor as CO_2 and SO_2 -rich slurry. The clean gas ($\text{CO}_2 \ll 1\%$) is then used for regenerating the dryers before being released to the atmosphere.

The pollutant-rich slurry (CO_2 & SO_2) from the Cryocontactor then gets pumped into the solid-liquid separator module (marked in blue). The filter press separates the now-solidified pollutants (CO_2 & SO_2) from the contact liquid. The contact liquid from the filter press recirculates back into the Cryocontactor, and the solid pollutants are melted and passed into the separation system module.

The separation system module (distillation panel) consists of a series of distillation columns designed to separate out CO_2 , SO_2 , and any traces of contact liquid. The separated high-purity CO_2 ($>99\%$) is in a state that can be easily transported without the need for any additional compression. The system is instrumented with TESTO to monitor the CO_2 concentration of inlet flue gas and exit clean gas.

Basis of Design

The below section describes details of the various parameters considered for a nominal capture of 1/4 ton of CO_2 capture/ day.

1. **Footprint:** The rig is mobile for ease of transport to multiple test sites. The rig is designed to fit over a 15 m trailer flatbed (Figure 2). The trailer includes a control cabin (air-conditioned), power cabinets and gas storage onboard. A diesel genset (with inbuilt fuel tank), to power the skid will be kept separate from this.
2. **Cryochiller technology:** This rig is built around a commercially available pulse tube cryochiller. For this design case, a cooling capacity of 2.2 kW is assumed at 110 K saturation temperature of Liquid Nitrogen that will be used as a refrigerant.
3. **Heat rejection:** The cryochiller has an option of integrated air-cooler to reject heat to the ambient. However, at higher ambient, this will impact the power requirement and cooling performance of the cryochiller. So, a separate water chiller is used on the skid to cool the water circulating through the cryochiller and the precoolers/aftercoolers on the skid. The total requirement for the water chiller is nominal 35 kW.
4. **Flue Gas Conditions:** The CCSC system is designed for flexibility, with a baseline suited to realistic operating conditions across various fuel types.
 - o Temperature: 170°C
 - o Pressure: -55 mbarg
 - o Additional pressure drop of -20 mbarg is assumed for a 2" 100m hose connecting the stack port to the skid.
5. **Flue Gas Composition:** Design point concentrations are summarized below
6. **Co-Capture:** SO_2 is present along with CO_2 in the system, the majority of the portion ($\sim 75\%$ SO_2) is removed along with CO_2 and the rest is removed before CO_2 separates out. Figure 4 shows the percentage of SO_2 and CO_2 that is co-captured in the CCSC rig.
7. **Site Conditions:** Using ambient conditions based on demonstration rig location at KAUST (Figure 5: <https://www.worldweatheronline.com/thuwal-weather-averages/makkah/sa.aspx>)

- Elevation above sea level: 0 m
- Ambient Temperature: 35°C
- Relative Humidity: 55%

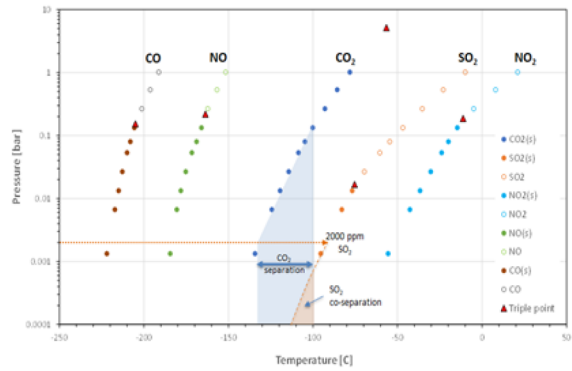


Figure 4—CO₂ & SO₂ Co-Capture

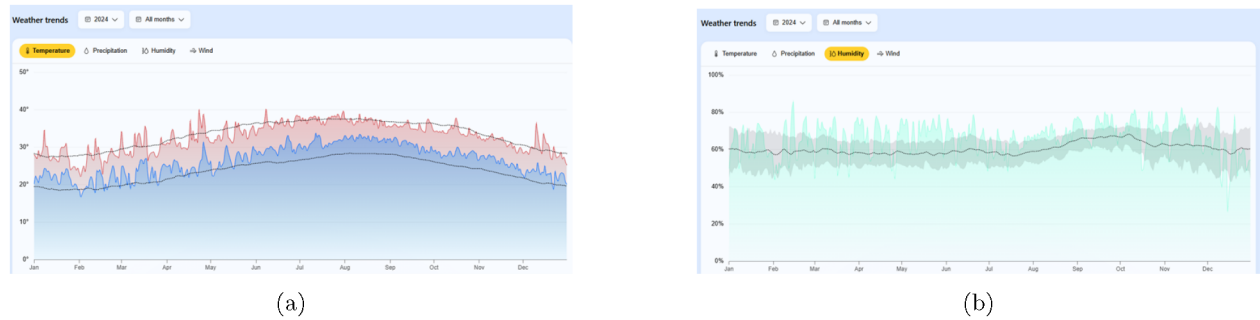


Figure 5—Yearly variation of temperature and humidity at Thuwal, Saudi Arabia

Component Species	Units	Value	Source
Carbon Dioxide	vol%	15	Dry Basis
Sulphur Oxides	ppm	2000	Dry Basis
Nitrogen Oxides	ppm	300	Dry Basis
Carbon Monoxide	ppm	330	Dry Basis

Safety Considerations

A general risk identification matrix was developed to assess hazards based on specific work areas within the trailer rig. The matrix outlines 30 potential risks identified through a comprehensive review of trailer-mounted operations. Following a detailed field inspection of the CCSC rig, each risk was evaluated and mapped to the corresponding work area or process module where it could realistically occur. This structured approach enabled the development of targeted mitigation strategies for high-priority zones and activities, ensuring safe operation during both field testing and routine use.

In parallel with the general risk assessment, a detailed evaluation of chemical-related hazards was conducted. The operation of the CCSC rig involves the use of various process chemicals, refrigerants, and byproducts—many under cryogenic and pressurized conditions. While individually these substances

are common in industrial settings, their combination within a confined, trailer-mounted system introduces complex safety considerations. To address this, a chemical hazard assessment was performed, focusing on toxicity, flammability, and physical hazards. The following section summarizes the key chemicals used in the CCSC process, categorized by the type of hazard they present and the corresponding mitigation measures implemented to ensure safe handling and operation.

Toxic & Asphyxiant Hazards

- Flue Gas contains toxic components and acts as an asphyxiant. It may be present at both low and high temperatures. Mitigation includes the use of pressure relief devices, thermal insulation, active ventilation, and routing gas relief to safe outdoor areas.
- Separated SO_x and NO_x are toxic and handled as condensed liquids at cryogenic temperatures. SO_x is safely neutralized via seawater discharge, while both gases are confined in closed systems with designated gas relief zones. Area ventilation and thermal insulation are applied to prevent accidental release and exposure.
- CO₂ is non-toxic but poses an asphyxiation hazard at high concentrations and is stored under cryogenic conditions. Mitigation measures include thermal insulation, ventilation, pressure relief devices, and continuous CO₂ monitoring.
- Nitrogen (gas or liquid) presents an asphyxiation risk due to oxygen displacement in confined spaces. It is managed using thermal insulation, pressure relief devices, and area ventilation.

Flammable Hazards

- Iso-pentane is highly flammable but non-toxic. It is stored under low-temperature, normal-pressure conditions. Risk is mitigated by the use of pressure relief devices, thermal insulation, explosion-proof design, and real-time monitoring with lower explosive limit (LEL) detectors.

Physical and Handling Hazards

- Perlite, along with high-performance cryogenic insulation sheets, is used for insulation within the cold box and is non-toxic and non-flammable, but it can irritate the eyes and lungs if dispersed. It is handled within an encapsulated system. PPE such as dust masks and goggles is mandated during loading or maintenance operations.
- Molecular Sieve Desiccant is chemically stable and safe, but poses a desiccation and slip hazard. It is housed in encapsulated cartridges, and the system is designed for easy access during replacement or maintenance.
- Water, used in the process, poses no chemical or physical hazard and requires no mitigation beyond standard handling.

Given the use of iso-pentane as contact liquid, it was essential to evaluate the potential for hazardous explosive atmospheres under both normal and off-nominal conditions. To this end, a detailed ATEX risk assessment was carried out for the CCSC rig, concluding that under normal operations, the potential for explosive atmospheres is minimal due to cryogenic containment and positive pressure control of iso-pentane, limiting Zone 2 areas to small localized regions. Additionally, a HAZOP identified an overflow scenario into non-ATEX components and was deemed physically improbable based on the system's layout and operational safeguards.

In addition to chemical exposure risks, the trailer-mounted CCSC system also presents several mechanical hazards due to its operation under cryogenic and pressurized conditions. The integration of various rotating equipment, thermal zones, and electrical systems within a compact layout demands a comprehensive evaluation of physical risks to personnel and equipment. The following section outlines the key mechanical

hazards identified and the corresponding mitigation strategies incorporated into the rig design and operation procedures.

High Pressure

Multiple pressure zones exist in the system, including approximately 1 bar, 8 bar, and 15 bar (absolute). Pressure buildup may also result from the sublimation or evaporation of condensed gases. The system has been designed to minimize potential gas entrapment, and pressure relief valves have been installed on all relevant lines. All installations are subjected to hydrostatic and leak testing after construction.

Noise

High-decibel noise from compressors and other equipment can pose risks to hearing and operational safety. Hearing protection (earmuffs) is required for all personnel working in high-noise areas as part of standard PPE.

Collision Hazards

The layout of the trailer rig allows adequate clearance around all skids for safe movement. All operations are monitored from a dedicated control room. Equipment such as valves and sensors is positioned within safe, accessible reach, avoiding the need for hazardous manual access.

Extreme Temperatures

The system operates across a broad temperature range—from cryogenic levels (-196°C) to high-temperature zones (up to 250°C). All exposed surfaces are insulated to prevent accidental contact. Operators are instructed to verify temperatures before interacting with equipment.

Vacuum Conditions

Vacuum formation can result from rapid cooling or suction in specific process sections. The system includes pressure monitoring and automated controls. Nitrogen injection is used as a safeguard to prevent vacuum conditions where applicable.

Rotating Equipment

All rotating equipment is fully enclosed to eliminate the risk of operator contact. Protective covers remain in place during operation and maintenance.

Electrical Hazards

Electrical systems are designed to prevent accidental exposure. All wiring and enclosures are compliant with industrial safety codes and standards.

Ventilation

Proper ventilation is critical due to the use of cryogenic fluids and inert gases. In normal operation, flap doors in the operational area must remain open to allow for adequate airflow and to prevent the buildup of hazardous gases. During transport, these doors are kept closed to protect equipment and maintain containment. Only the monitoring and storage rooms may remain enclosed during operation, as no gas lines are routed through the monitoring room, minimizing leak risk in this space. To further ensure safety, operators are required to carry portable O₂ detectors, particularly during startup and connection activities, when the potential for oxygen displacement is highest.

Test Campaign and Results

The CCSC rig was deployed and tested at the Saudi Electricity Company's largest HFO-fired power plant. The test campaign was structured around several key phases:

- **Pre-Commissioning:** This included site preparation, tie-in connections readiness, comprehensive risk assessments, and the development of Job Safe Work Procedures (JSWP).
- **Mobilization and Commissioning:** The trailer rig was transported, positioned, and mechanically/electrically integrated. Functional testing and dry runs were conducted to verify system readiness.
- **Pre-Startup Safety Check (PSSC):** A safety review was conducted to confirm equipment integrity, hazard controls, and readiness for live operation.
- **Stack Tests:** Actual flue gas from the power plant was introduced into the system to capture CO₂, SO₂, and NO₂. The process was actively monitored and optimized in real-time to ensure peak capture performance.
- **Decommissioning and Site Restoration:** Following successful testing, the system was shut down as per operating procedures. Equipment was demobilized, and the site was restored to its original state as per host requirements.

Following the completion of major activities, including site preparation, system tie-ins, risk assessments, mobilization, commissioning, and pre-startup safety checks, the CCSC rig was ready for field testing. A series of stack tests was then conducted to evaluate system performance under both simulated and real flue gas conditions. The following section summarizes key outcomes from these tests, demonstrating the capture efficiency and validating the robustness of the CCSC process.

Simulated Gas Composition

Prior to injecting real flue gas, simulated gas tests were conducted to evaluate capture efficiency at different operating temperatures and CO₂ concentrations. Figure 6 shows CO₂ capture efficiency versus setpoint temperature for exhaust streams containing 15% (blue dotted) and 50% (red dotted) CO₂. Among multiple simulations conducted, two representative operating points are highlighted in this paper: at a setpoint of -111°C for the 50% CO₂ case, the thermodynamic model predicts a capture efficiency of 95%, whereas at a slightly warmer setpoint of -105 °C for the 15% CO₂ case, the capture efficiency is 48%. The intersection of each operating point with the respective curve demonstrates the sensitivity of capture efficiency to both inlet CO₂ concentration and temperature control.

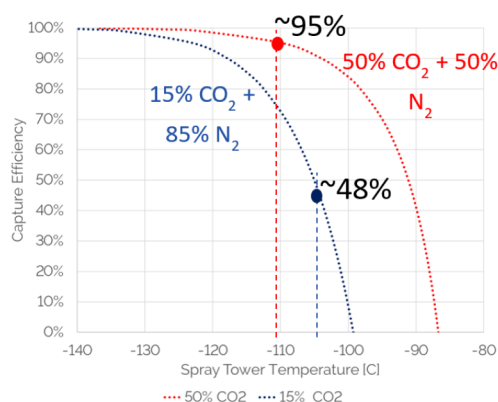


Figure 6—CO₂ capture efficiency as a function of system setpoint temperature for simulated exhaust streams

Figure 7 shows measured CO₂ capture performance for the two simulated exhaust compositions discussed previously. In the 15% CO₂ case, the model predicted a capture efficiency of 48%, whereas the measured value was 44%. The plotted traces correspond to CO₂ concentrations at two sample points on the trailer rig: the black solid line represents the inlet stream to the CCSC system, and the orange solid line represents the outlet stream. Although the x-axis (time) is not shown, the test was conducted over

extended continuous operation, during which the outlet concentration exhibited stable, asymptotic behavior. The deviation between measured and predicted capture efficiency can be attributed to the steep gradient of the capture- efficiency-versus-temperature curve at this operating point, which amplifies the effect of small temperature deviations. Additional contributing factors may include: (i) sensor uncertainty for prolonged exposure to CO₂ (intermittent rinsing needed), (ii) transient temperature mixing effects within the cryo-contactor

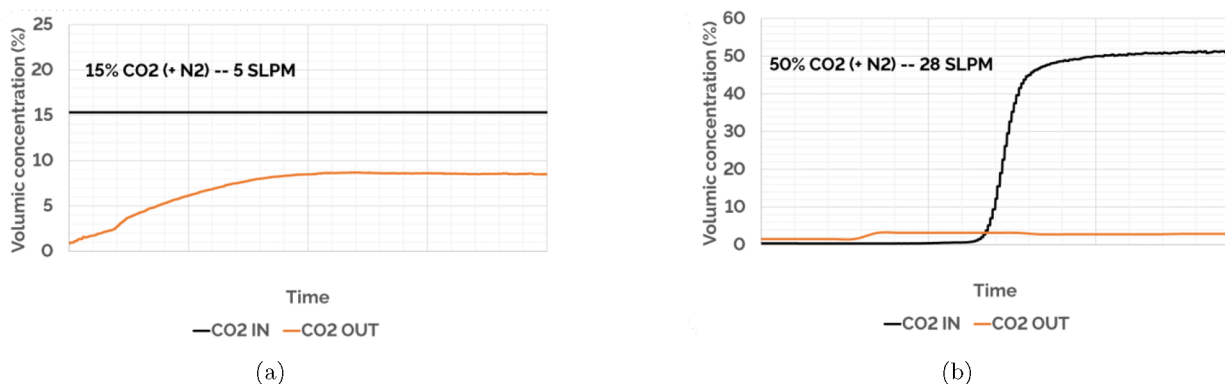


Figure 7—CO₂ Capture Performance: Inlet vs. Outlet Concentration Trends

In the 50% CO₂ case, the model predicted a capture efficiency of 95%, with measured performance of 94%. The reduced discrepancy here reflects the flatter slope of the capture-efficiency curve at the selected setpoint temperature, making the system less sensitive to small process variations.

These results validated the accuracy of our process model and built confidence ahead of full-scale testing with actual flue gas.

Flue Gas Injection

Prior to injecting real flue gas, the system was stabilized and cooled down to its designated operating set point of -130°C . The flue gas, among other pollutants, had a CO₂ concentration of approximately 15% (dry basis). The expected capture efficiency for this temperature and CO₂ concentration is theoretically greater than 95%.

Actual flue gas injection tests were performed at a setpoint temperature of $-130^{\circ}\text{C} \pm 3^{\circ}\text{C}$. The primary y-axis in Figure 8 indicates measured CO₂ concentration, and the secondary y-axis presents the derived capture efficiency based on inlet and outlet measurements. The x-axis shows elapsed time, allowing assessment of performance stability throughout the test period.

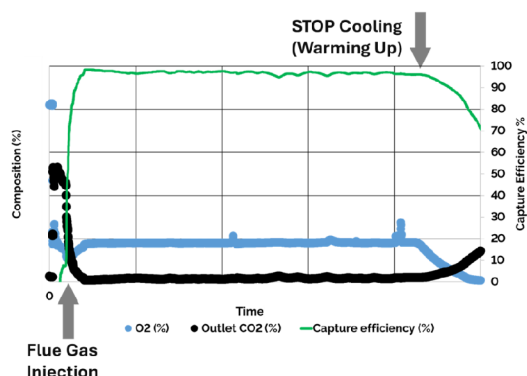


Figure 8—CO₂ Concentration and Capture Efficiency During Flue Gas Injection

Once the flue gas injection commenced, the outlet CO₂ concentration, measured using a Testo gas analyzer, immediately dropped, indicating effective capture. During extended periods of operation, Figure 8 shows that the capture efficiencies consistently exceeded 95%, aligning closely with theoretical predictions. In parallel, oxygen concentration was monitored throughout the process. Despite the presence of O₂, the system operated well below the Lower Explosive Limit (LEL) of iso-pentane due to cryogenic conditions and positive pressure operation, thus eliminating explosion risk.

The corresponding graph demonstrates the CO₂ and O₂ trends during testing. The data has been intentionally sanitized to protect proprietary project information; however, it remains representative of actual system behavior and validates the effectiveness of the CCSC capture process.

Notably, the system also achieved near-zero SO₂ and NO_x levels at the outlet stream (not shown here), using a single integrated cryogenic technology that consolidates FGD and SCR functionalities. This results in substantial CAPEX and OPEX savings while streamlining post-combustion gas treatment.

Summary

The CCSC system demonstrated effective CO₂ capture efficiency of over 95% during real flue gas injection and achieved near-zero SO₂ and NO_x emissions (from 1700 ppm), validating its performance. These tests were witnessed by all key stakeholders, reinforcing confidence in the controls and operating philosophy.

By integrating the functionalities of FGD and SCR into a single cryogenic platform, the system offers significant CAPEX and OPEX advantages while simplifying post-combustion treatment. Designed for minimal disruption to live power plant operations and aligned with industry best practices, the system also recovers usable water from gas streams with minimal operational water input.

This cryogenic approach shows promise in replacing conventional SO_x and NO_x control systems, currently a focus of ongoing research at KAUST in partnership with SEC, and potentially further enhanced through integration with energy storage to support renewable energy deployment.

Acknowledgements

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